

Dr. Babasaheb Ambedkar Technological University, Lonere
Summer Examination May 2015

B.Tech Course in Electronics and Telecommunication Engineering
Subject : Principle of Communication Engineering
Semester: IV

Date: 16/05/2015

Time: 3 Hours

Max Marks : 70

Instructions to the students

1. Q1 is compulsory and carry 10 marks whereas Q2 to Q7 carries 12 marks each.
2. Attempt any five questions from Q2 to Q7.
3. Illustrate your answers with neat sketches, diagrams etc. whenever necessary.
4. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is a part of examination.
5. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.
6. Zero marks have been given for multiple answers to the chosen question.

Q1. Select an appropriate option for each of the following

[10]

- i) A dBm measurement for audio input and output specifications, the specified reference level is
 - a) .001W
 - b) 1W
 - c) 1V
 - d) 1mV
- ii) Thermal noise is independent of
 - a) bandwidth
 - b) temperature
 - c) centre frequency
 - d) Boltzmann's constant
- iii) Communication systems are often categorized by the
 - a) frequency of message signal
 - b) frequency of carrier signal
 - c) band width of the channel
 - d) None of the above.
- iv) The total area under PDF is
 - a) 0
 - b) 1
 - c) depends on the nature of the distribution
 - d) ∞
- v) The density function of a random variable X is given by, $f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$. The variable's distribution is
 - a) Poisson distribution
 - b) Gaussian distribution
 - c) Uniform distribution
 - d) Rayleigh distribution
- vi) The image channel rejection in a super-heterodyne receiver comes from
 - a) IF stage only
 - b) RF stage only
 - c) detector & RF stage only
 - d) detector, RF and IF stages
- vii) The resonant frequency of a RF amplifier is 1MHz; its bandwidth is 10 kHz. The Q-factor will be
 - a) 10
 - b) 100
 - c) 0.01
 - d) 0.1
- viii) The modulation index of an AM wave is changed from 0 to 1. The transmitted power is
 - a) Increased by 50%
 - b) unchanged
 - c) halved
 - d) quadrupled
- ix) In PCM, if the number of quantization levels is increased from 4 to 64, then the bandwidth requirement will approximately be
 - a) 3 times
 - b) 16 times
 - c) 4 times
 - d) 8 times

- x) In a modulator, it is found that the amplitude spectrum consists of a carrier frequency component and two sidebands. The modulator used is
- a) AM b) AM and/or NBFM c) FM d) PAM

Q2. Attempt any TWO of the followings

- a) A microwave transmitter typically requires a +8dBm audio level to fully drive the input. If +10dBm level is measured, what is the actual voltage level measured? Assume 600Ω system. [6]
- b) Consider the random process $X(t) = A \cos(\omega_c t + \theta)$ where A and ω_c are constant while θ is random variable with an uniform pdf $f_\theta(\theta) = \frac{1}{2\pi}$, $-\pi \leq \theta \leq +\pi$
- i) Find mean, autocorrelation function and show that $X(t)$ is WSS process.
- ii) Find PSD of $X(t)$. [6]
- c) Define modulation and explain briefly in following context with suitable example;
- i) Multiplexing ii) Design practical antenna iii) Narrowbanding [6]

Q3. Attempt the following

- a) Describe generation of DSB-SC signal using balance modulator. Draw spectrum for the balance modulator output signal and specifications of BPF to generate required signal. [6]
- b) A conventional AM signal $s(t) = A_c[1 + 0.8\cos 500\pi t]\cos 5000\pi t$ is passed through an ideal filter that removes the LSB. Obtain the expression for the resulting SSB-Large Carrier signal if it delivers a power of 2.32 W to a 1Ω resistor load. [6]

Q4. Attempt the following

- a) Describe double spotting in AM super-heterodyne receiver with appropriate example. [6]
- b) An AM broadcast receiver has an IF of 455 KHz and is tuned to 1200 KHz. The RF stage has a tuned circuit with Q of 80
- i) Obtain the image frequency
- ii) Find the image rejection in dB. [6]

Q5. Attempt any TWO of the following

- a) Design an Armstrong indirect FM transmitter operating at a carrier frequency of 96MHz with permissible frequency deviation in the range 80-85 KHz. The following may be assumed to be available
- i) An NBFM generator with carrier frequency 100KHz and peak frequency deviation 10Hz, and $\beta = 0.5$.
- ii) Any number of frequency doublers.
- iii) VFO in the range of 10-15MHz. [6]
- b) Given the single tone FM signal $s_{FM}(t) = 20\cos[2\pi \times 10^6 t + 2\sin(2\pi \times 10^4 t)]$
- i) Sketch the FM spectrum for the carrier and first three sideband terms.
- ii) What is bandwidth using Carson's rule.
- Assume: Normalized side band component as: $A_c J_n(\beta)/2$
 $J_0(2) = 0.224$, $J_1(2) = 0.577$, $J_2(2) = 0.351$, $J_3(2) = 0.129$ [6]
- c) Describe in detail how a PLL can be used to detect FM signal. [6]

Q6. Attempt any THREE of the following

- a) What is quadrature null effect and explain one application based on it. [4]

- b) Show that figure of merit, FOM is one in SSB-SC receiver [4]
- c) Let PSD of AWGN be $N_0/2$ Watt/Hz. If it is passed through a system, whose transfer function $H(f)$ is $2e^{-j2\pi ft}$, followed by an ideal LPF with bandwidth B Hz. Calculate output noise power at the output of LPF in watts. [4]
- d) 1mW video signal having bandwidth 100MHz is transmitted to a receiver through cable that has 40 dB losses. If effective one sided noise PSD in receiver is 10^{-20} Watt/Hz, calculate SNR at receiver input. [4]
7. Attempt the following
- a) 4 voice signals each band limited to 4 KHz, sampled at Nyquist rate and converted into PCM signal using 256 levels. Calculate band width and bit rate of the PCM signal. [6]
- b) Explain in detail PAM technique and mention one application of analog pulse modulation technique. [6]